



# RAMCO INSTITUTE OF TECHNOLOGY

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## Department of Computer Science and Engineering Academic Year 2022 – 2023 (Even Semester)

**Degree, Semester & Branch: II Semester B.E. CSE 'A'**

**Course Code & Title: CS3251 Programming in C**

**Name of the Faculty member (s): Mrs.B.Vijayalakshmi, AP (SG)/CSE**

### Innovative Practice Description

- **Unit / Topic: Unit I / Looping Statement**

- **Course Outcome: CO1**

- **Topic Learning Outcome: TLO3**

- **Activity Chosen: Code Debugging**

- **Justification:**

Strings are the important concept which has many built in methods for processing. In order to remember and recollect how to apply these methods and also to familiarize the syntax, a code completion activity is given to the students.

- **Time Allotted for the Activity: 15 Minutes**

- **Details of the Implementation:**

- Students were asked to form a team of two members for this activity. Different set of questions were designed.
- In this activity, each team was given with a C code snippet involving the various kinds of looping statements in C in which partially completed code along with the desired output is given. The students have to fill the incomplete code and debug the code to remove the errors from them for getting the desired output.
- The sample questions and the student's answers are shown in Fig 1.
- The activity was conducted during the last 15 minutes of the class. In the first 10 minutes the students have to think and identify the missing code and debug it to get the desired output.
- At the end of the activity, the answers for the questions were discussed by the student team members.

- **CO – PO / PSO mapping:**

| CO  | PO1 | PO2 | PO3 | PO5 | PO8 | PO9 | PO10 | PO12 | PSO1 | PSO2 | PSO3 |
|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 3   | 2   | 1   | 2   | 1   | 1   | 1    | 1    | 3    | 1    | 1    |

(1 – Low      2 – Moderate      3 – High)

- **PO / PSO mapped:**

| Innovative practice | PO1 | PO2 | PO3 | PO9 | PO10 | PSO1 | PSO2 | PSO3 |
|---------------------|-----|-----|-----|-----|------|------|------|------|
|                     | 3   | 2   | 1   | 1   | 1    | 3    | 1    | 1    |

|                                      |                                                       |                                              |                                                                       |                                                                 |                                                      |                                                |                                              |                                               |
|--------------------------------------|-------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|----------------------------------------------|-----------------------------------------------|
| <b>Justification for correlation</b> | The basic knowledge about looping statement is needed | The students analyze the given partial code. | The students debug the code to answer for the given coding questions. | The students coordinate and demonstrate the activity as a team. | The students communicate the solutions to the others | The concept is used in developing the software | The concept is used in security applications | The concept is used in solving problems in AI |
|--------------------------------------|-------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|----------------------------------------------|-----------------------------------------------|

• **Images / Screenshot of the practice:**

Identify the error and complete the code to get the desired output.

```
#include <stdio.h>
int main()
{
    char n, i; int n, i = 0;
    printf("Enter an integer: ");
    scanf("%d", &n);
    for (i = 0; i <= 5; ++i)
    {
        printf("%d * %d = %d \n", n, i, n * i);
    }
    return 0;
}
```

**Output:**

Enter an integer: 7

7 \* 1 = 7  
7 \* 2 = 14  
7 \* 3 = 21  
7 \* 4 = 28  
7 \* 5 = 35

Akathesh Sri Ram  
Smana Yogeswar -

Identify the error and complete the code to get the desired output.

```
#include <stdio.h>
int main()
{
    int num, i;
    printf("Enter a positive integer: ");
    scanf("%d", &num);
    printf("Factors of %d are: ", num);
    for (i = 1; i <= num; i++)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
    return 0;
}
```

**Output:**

Enter a positive integer: 10  
Factors of 10 are: 1 2 5 10

Thirumoorthi. R  
Aravind. A. S

**Fig 1: Students Response**

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- The students feel it is very interesting to debug and complete the partial code. They can easily identify the syntax errors they will frequently encounter during coding.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
  - It helps the students to remember the syntax and the working of looping statements.
- ❖ ***Challenges faced in implementation:***
  - Some of the students those who does not understand the looping statement clearly felt difficult to find answers.

**References:**

- ❖ [https://www.ritrjpm.ac.in/images/computer-science/26\\_OIT551\\_Colloborativecoding.pdf](https://www.ritrjpm.ac.in/images/computer-science/26_OIT551_Colloborativecoding.pdf)
- ❖ [https://www.ritrjpm.ac.in/images/computer-science/18\\_EC8393\\_Codedebug.pdf](https://www.ritrjpm.ac.in/images/computer-science/18_EC8393_Codedebug.pdf)